

Chapter 35

The Futures of Globalization

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INTRODUCTION: DEFINING THE FIELD

Theories of globalization have been the dominant paradigm in sociology for at least two decades, but certain features of the globalization debate have been part of sociological discourse for much longer. In mainstream academic sociology, one of the earliest publications on the topic was by W.E. Moore (1966) in his 'Global sociology: The world as a singular system'. He argued that sociology was becoming a global science and that 'the life of the individual anywhere is affected by events and processes everywhere' (Moore 1966: 482). 'Globalization' refers then to the process by which the world becomes a single place, and hence the volume and depth of social interconnectedness are greatly increased. Globalization is the compression of social space (Giddens 1990). The analysis of the future of globalization will have to address the consequences, both intended and more commonly unintended, of these processes of temporal and spatial compression.

In the 1960s Marshall McLuhan (1967) had introduced an influential vocabulary to describe the role of 'the global village' in the analysis of culture and mass media in order to understand how the world was shrinking as a result of new technologies of communication. The globalization literature grew apace in the 1970s and 1980s, mainly within the sociology of religion where religious revivalism was increasingly seen as a global process (Beckford and Luckmann 1989; Robertson 1987). By the 1990s globalization had been identified as 'the central concept' of sociology (Robertson 1990).

Although there is now a large and expanding body of literature on globalization, theories of these global processes are characterized by certain persistent deficiencies, especially in terms of explanatory power and precision. This discussion is concerned with the future of globalization processes rather than with the future of globalization theory, but of course the two issues are inevitably interconnected. If we are to think effectively about globalization's futures, it is helpful intellectually to consider briefly some of the difficulties and shortcomings of existing theories in the

sociological literature. In this commentary, the range of the debate about the nature of globalization is extended through a sociological interpretation of its likely futures. Because futurology is generally speaking merely an extrapolation from present trends, many of these social transformations are of course already upon us.

Religious dimensions of globalization have been neglected, and most explanations focus broadly on technological and economic causes (Beyer 1994; Turner 1994). For example, while Ulrich Beck (2000: 53) clearly recognizes the importance of cultural globalization and 'ideoscapes', *What Is Globalization?* contains no discussion of fundamentalism, Islamic radicalism or religion in general. Sociologists have in addition had little to say about military globalization or about warfare. The impact of war and militarism on the origins and development of globalization has thus been neglected (Black 1998), and yet they have played a crucial part, especially with the rise of world wars, in transforming the international order into a global system. In the globalization literature, there has developed an unfortunate gap between sociological and international relations theory. Religion and military violence are therefore important causal aspects of globalization processes.

In mainstream sociology, the most influential writer on the importance of religion in globalization has been Roland Robertson (1992), who complained with some justification that analysts had overstated the economic nature of globalization (free trade, neoliberalism, financial deregulation, and integrated production and management systems) to the neglect of its social and cultural characteristics, especially its religious dimensions. Theories that emphasize the technological and economic causes of globalization (such as computerization of information and communication or economic and fiscal deregulation in the neoliberal revolution of the 1970s) show little appreciation of long-term cultural, religious and social conditions. These theories tend to be somewhat simple versions of technological determinism. Whereas Ulrich Beck (1992) and Anthony Giddens (1990) have approached globalization as an aspect of late modernity (and therefore as related to risk society and reflexive modernization), Robertson has been concerned with long-term cultural developments. These include the unification of global time, the spread of the Gregorian calendar, the rise of world religions, the growth of human rights values and institutions and the globalization of sport. In short, we also need to attend to the various dimensions of globalization and their causal priority: such dimensions as the economic and technological (including global markets of goods, services and labour); the informational and cultural (such as global knowledge, religious revival movements and radical fundamentalism); the legal and political (human rights, legal pluralism and legal regulation of trade); and the medical and health aspects (such as epidemics). We can simplify this discussion by suggesting that globalization has four major dimensions: economic, cultural, technological and political. Any comprehensive analysis of the futures of globalization would have to consider all four dimensions and their interaction, but this overview has of necessity to be more selective in its treatment of issues. Furthermore, this analysis will be primarily concerned with the negative and unintended consequences of these four broad aspects of global society.

Perhaps one consistent weakness of sociological views of globalization has been their persistently optimistic, and frequently naive assumptions about the causes and consequences of globalization. There are of course existing criticisms of

globalization that have identified its negative features. For example, it has been claimed that globalization is basically the spread of rational models of organization that produce cultural standardization. McDonaldisation is the classic example (Ritzer 1993). Globalization produces a standardized world, not simply of commodities, but also of traditions and cultures. Globalization empties out social content, evaporates cultural differences and eliminates authenticity. The result is cultural nothingness (Ritzer 2004). Anti-globalization theorists and activists who obviously share this view oppose globalization as the imposition of Western economic hegemony. Beck has associated globalization with risk society, thereby exploring its negative impact on social structures and everyday life. These existing criticisms of globalization theory have been brought into sharper focus by the recent interest in 'empire' (Hardt and Negri 2000) and the question of global military conflict is therefore closely allied to arguments about the rise of an American empire. Niall Ferguson (2004) in *Colossus* claims that American imperialism has a long history, but there is a persistent sense that imperialism and democracy do not go together, and hence America is an empire in denial. Recent studies of emerging imperial power suggest that global governance requires an imperial policeman to regulate local conflicts and regional wars. If Ferguson is correct, globalization will require new systems of regulation and domination which cannot be provided by nation-states.

While there are important criticisms of globalization (as process rather than merely as theory) in the evolving literature on globalization, the attack on the World Trade Center, the subsequent 'war on terrorism' and more recently the bombings in Madrid and London have transformed the study of globalization. International terrorism has, at least for the foreseeable future, eclipsed erstwhile naively optimistic views of globalization as merely a positive extension of modernization. There is currently therefore a greater interest in the negative aspects of globalization, namely of urban violence, slavery, terrorism, international crime, tourist sex and global epidemics. In response to this negative but more realistic assessment of globalization, I develop what can be regarded as a neo-Malthusian theory that examines the complex interaction between the clash of civilizations, terrorism, changing population structure, ageing and migration, pollution and environmental crisis.

The futures of globalization will require the colonial and imperial exploitation of outer space, because the earth's resources will be too depleted to support future population growth. If the earth's resources are finite, then space exploration must have a significant economic and military role in the development of global society. This dystopic vision was perhaps first articulated by H.G. Wells in his *The Time Machine* which was a major contribution to futurology. Wells also played a significant role in the development of human rights when in a series of letters to *The Times* in October 1939 he helped to create a bill of rights against totalitarian regimes (Robertson 2002). The future of globalization will involve a significant increase in the disruption of civil society through 'new wars', and a variety of attempts to secure the rule of law, including the globalization of rights. These civil conflicts will have diverse manifestations, but a common set of causes.

This analysis of the futures of globalization has the following general features. First, it draws on the work of Michel Foucault to place the body and populations at the centre of any picture of the future, and hence this discussion will deal extensively with medical technology, population changes, ageing demographic structures,

disease and drugs (Turner 2004). One problem with Foucault's theory was the lack of a genuine political economy, namely a concern for governmentality in terms of pension funds, investment in regenerative medicine, research budgets for stem cell development and so forth. By combining Foucault and the political economy of ageing and resources, it is possible to develop a general framework for a neo-Malthusian sociology of globalization that attempts to grasp the interconnections between the following conditions: waste, environmental damage and the depletion of natural resources; the growth of radical politics, new wars and the alienation of youth; the growing importance of human rights responses to failed states and civil war; and the centrality of religious imagination and religious violence to global change. The aim is to understand how the changing structure of populations has interacted with capitalist markets to produce a new age of violence, and hence a key component of the future of globalization must be to consider the emergence of dystopic cosmologies, that will resemble the depraved consciousness of plague-ridden medieval society or the extraordinary baroque mentality of the Post-Reformation world.

CULTURAL GLOBALIZATION: RELIGION AND THE CLASH OF CIVILIZATIONS

The modern economy requires considerable labour mobility, and hence it requires open, porous state boundaries. The globalization of the labour market creates cultural diversity, and cultural diversity has become essentially religious diversity. Against this rising hybridity, the nation-state has sought to protect its sovereignty and to retain a largely homogeneous civil society. There is therefore a tension between the global economy and the territoriality of the nation-state. One traditional solution has been to develop multicultural policies, but for various reasons multiculturalism is under attack (Glazer 1997). It is, for example, largely incompatible with religious fundamentalism, because fundamentalist theologians recognize that such hybridity erodes the purity (that is the fundamentals) of religious orthodoxy.

One perspective on global hybridity emphasizes the growing clash of civilizations in which the conflict between religions defines the boundaries of the political. Samuel Huntington's article on 'the clash of civilizations' in *Foreign Affairs* (1993) has shaped much of the academic debate about inter-cultural understanding for almost a decade. In the post-9/11 world, Huntington's bleak analysis of the development of micro fault-line conflicts and macro core state conflicts has captured the mood of foreign policy in the West in the era of the 'war on terror'. Huntington of course believes that the major division is between the Christian West and the Muslim world. More recently, he has even more openly spoken about 'the age of Muslim Wars' and widespread Muslim grievance and hostility towards the United States (Huntington 1996). Any attempt to engage with Islamic civilization is now seen as a 'war for Muslim minds'. Globalization will continue to have contradictory consequences for religion: increasing pluralism and hybridity, growing fundamentalism and evolving importance of religious idioms as political ideologies. Globalization has had the further paradoxical effect of making religions (via their religious leaders and elites) more self-conscious of themselves as 'world religions', that is 'modern

conditions have made religions more self-consciously global in character' (Smart 1989: 556). We can trace this development of a global consciousness of a system of religions back, for example, to the World's Parliament of Religions in 1893.

The development of religious pluralism or the emergence of a religious super-market represents a consumer secularism in which religious forms are hybrid. Globalization produces cultural hybridization, because the interaction of different cultures through migration and the growth of diasporic communities creates, primarily in global cities, a shared cultural complexity (Appadurai 2003). These hybrid forms of religion are often constructed self-consciously and they are closely related to youth movements and to generational politics. Global, hybrid religiosity is a form of religious popular culture. In the United States, sociologists have identified the emergence of a 'quest culture' that attempts to find meaning in various and diverse traditions. The result is religious hybridity. The mechanism by which these hybrid religious styles emerge is through a 'spiritual market place' (Roof 1993, 1999). These quest cultures have been critically evaluated as forms of expressive individualism, because they are related to what Talcott Parsons called the 'expressive revolution' (Parsons 1974). New age communities have become a popular topic of sociological research (Heelas 1996), but we need to understand more precisely how these spiritual marketplaces function globally and how they are connected electronically through the Net (Dawson and Henneby 2003).

Sociologists have in the debate about religious globalization narrowly focused on fundamentalism. Indeed, the growth of religious fundamentalism is often regarded as the principal consequence of globalization (as the most recent aspect of modernization). Fundamentalism has been a common development in the three Abrahamic religions, but it is also present in various reform movements in Buddhism (Marty and Appleby 1995). Religious fundamentalism is often defined as traditionalism because it is seen to be anti-modern. This view pervades Frank Lechner and John Boli's *The Globalization Reader* (2004: 326–60). There is obviously evidence to show how fundamentalism has attempted to contain the growth of cultural hybridization, to preserve what is seen to be the pristine, authentic faith, to sustain religious authority and orthodoxy, and in particular to curb the growth of women's social and political autonomy. This view is questionable for two main reasons. First, fundamentalist movements employ the full range of modern means of communication and organization and secondly they are specifically anti-traditionalist in rejecting the taken-for-granted assumptions of traditional practice (Turner 2003). In Islam, since the formation of the Muslim Brothers in the 1920s, fundamentalists have consistently rejected traditional religion, specifically Sufism. If we consider fundamentalism as a verb, 'to fundamentalize', then religious traditions can be seen as strategies of alternative modernity, not a return to tradition (Turner 2002).

The growth of higher education and student movements has been important in the evolution of 'political religions'. The case of Islamic fundamentalism and political Islam is the classic illustration. With the decline of communism, radical religion has replaced secular politics as the rallying point of those who have experienced social alienation. It is of course important to distinguish between the training of different social strata with respect to universities, technical colleges and religious institutions. This argument – religion as an expression of the social disappointment and dislocation of economic change – is now inevitably associated with the

Huntington thesis and with the 'Jihad vs. McWorld' dichotomy (Barber 2001). The process of 'securitization' in Europe and North America and the development of the friend/foe approach to foreign policy has produced significant forms of political 'blowback'. Although the clash of civilizations thesis has been the subject of bitter criticism, it is evident that fundamentalist religions are incompatible with cultural diversity, legal pluralism and multiculturalism. Fundamentalism is generally hostile to inter-cultural marriage, secular education, tolerance for 'deviant' sexuality and women's involvement in the public sphere, that is, it is hostile to the conditions that make multicultural global cities feasible as spaces to inhabit. High rates of intermarriage between separate communities are a condition of successful social integration in multicultural societies, but fundamentalism undermines such conditions. In July 2005 the Muslim Council of Indonesia issued legal pronouncements (*fatwa*) banning mixed marriages, interfaith prayers and religious pluralism. The hybridization of cultures that is one consequence of globalization is not compatible with the defence of religious exclusivity.

TECHNO-MILITARY GLOBALIZATION: TERRORISM AND NEW WARS

The impact of military technology, militarism and war on the shape and direction of globalization has been neglected in the primary literature on globalization. The principal exceptions are *War and Modernity* (Joas 2003) and *War and Power in the 21st Century* (Hirst 2001). War and military technology are major causes of globalization because they determined the conditions of social development in the twentieth century. The spread of nuclear weapons shaped the conditions of warfare and produced the Cold War. Constant innovation in military technology shaped the economic growth of the Western world. However, it is said that there has been a revolution in military affairs (Cohen 1966), because the military platforms of existing forces – tanks, planes and ships – are obsolete. In the recent sociology of the military, there has been an important debate about the distinction between old and new wars, that provides a valuable insight into ethnic-cleansing and genocide. In particular, the concept of new wars is helpful in thinking about the increased vulnerability of women in civil conflicts. Military conflict has the effect of disrupting and eroding political boundaries, and 'new wars' will be an increasing feature of violence under global conditions (Münkler 2005).

Old wars or 'bourgeois wars' are said to be characteristic of the international system that was created by the Treaty of Westphalia of 1648, involving military conflict between armies that were recruited and trained by nation-states. In conventional inter-states wars involving large set battles and military manoeuvres, sexual violence against women on enemy territory was dysfunctional in terms of strategic, rational, military objectives. Attacks on civilian populations interfered with the primary objective of war, which was the decisive defeat of an opposing army by direct military engagement. Harassing civilian populations only constrained military mobility on the battle field and delayed direct engagement with an opposing army. Within these conventional inter-state wars, the development of international law to protect civilians was perfectly compatible with conventional military objectives. In

new wars, this military logic – its strategic rationality – evaporates, and systematic rape of women (so-called ‘camp rape’), and systematic violence towards civilians generally, become functional activities in the erosion of civil authorities and destruction of civil society. In bourgeois wars, rape was common but unplanned; in bourgeois wars, rape is systematic and constitutive of military violence. Therefore, in bourgeois wars, the majority of casualties were military personnel; in new wars, the casualties are almost entirely civilian. New wars involve the sexualization of violence, because civil society can be destroyed through a systematic attack on the reproduction of its population. Mass sexual violence has a symbolic political meaning, namely that men can no longer protect their women. Another characteristic of such wars is the growing use of children as cheap and biddable combat troops. These wars are in large measure the product of failed states and the reduced cost of military equipment, such as the widespread use of the Kalashnikov rifle. One consequence of new wars in Africa, especially in the Congo, has been growing distrust of and hostility towards children. Child soldiers have played a major role in civil conflicts throughout Africa and are no longer regarded as innocent victims of war. The extended family has largely broken down in central Africa leaving many children without care and support. In cities such as Kinshasa, there has been a sudden increase in witchcraft accusations against marginalized children. Indigenous evangelical churches play an important role in exorcising possessed children, and hence there is a causal connection between state failure, new wars and generational disruption. In Africa and Latin America, globalization is closely associated with religious revivalism such as Pentecostalism (Lehmann 1996; Martin 2002).

New wars have occurred and are occurring in Afghanistan, Bosnia, Chechnya, Myanmar, East Timor and the Sudan. In these wars, the military institutions and technologies of nation-states, especially of America, are largely irrelevant to the conduct of new wars, a situation which is illustrated by the inability of coalition forces to contain violence in Baghdad. Tanks are relatively useless against suicide bombers mingling with urban crowds on urban subways. New wars, not bourgeois wars, will characterize the future of globalization, because nuclear wars between nation-states cannot be successfully pursued without mutual destruction.

POLITICAL GLOBALIZATION: THE JURIDICAL REVOLUTION

Human rights legislation and institutions such as the international court of justice have been responses to the humanitarian crises of world wars. The ‘juridical revolution’ of the twentieth century, involving the international recognition of human rights as formulated in the United Nations Declaration, is a major example of the general process of legal globalization (Ignatieff 2002; Woodiwiss 2003).

There have been several stages in the development of human rights. In the final quarter of the eighteenth century, American independence from Britain and the French Revolution in overthrowing despotic monarchy established the liberty of the individual as a constraint on the powers of the state. Immanuel Kant defined ‘enlightenment’ as freedom from tutelage to think and act as an autonomous individual. Enlightenment ideas had a significant impact on the French Revolution, which proclaimed the famous ideas of liberty, equality and fraternity. The essential

principle behind the English, American and French revolutions was that human beings should not be subject to arbitrary and tyrannical powers.

The second stage was marked by the Nuremberg trials in which the Nazi crimes against the Jews came to be defined as 'crimes against humanity' and Article 6 (c) of the Nuremberg Charter established the legal grounds by which those responsible for torture and genocide would be prosecuted in an international court of justice. These human rights laws recognized individuals as victims of state crimes, and made special provisions for political refugees. Throughout the Cold War, human rights became part of the rhetorical struggle with the Soviet Union. International protection of refugees was an aspect of Cold War international relations, where acceptance of refugees was an aspect of this ideological struggle. With the fall of the Soviet Union in 1992, Western nations, supported by neoliberal economic strategies, came to regard refugees as a threat to national culture and economic growth. With the end of the Cold War, there is greater emphasis on asylum abuse, people trafficking, repatriation, burden sharing and short-term protection. The decline of universal norms regulating the flow of people on the part of states has seen a growth in voluntary-sector responses from international non-governmental organizations (INGOs) and non-governmental organizations (NGOs), especially in the care of women and children who constitute the overwhelming majority of refugees. Human rights abuse by states appears to be the main cause of displacement and forced migration.

In the late twentieth century, there has been an age of enforcement including truth commissions in post-Apartheid South Africa, the extradition of Pinochet, the occupation of East Timor by UN troops, the Lockerbie verdict, the Japanese apology for war-time treatment of 'comfort women' and the trial of Slobadan Milosevic. These legal and political efforts to enforce human rights conventions are seen to counteract the criticism that human rights are not justiciable without world government. However, the international failure to act effectively over the Darfur crisis in western Sudan in 2003–4 illustrates the argument that the UN is reluctant to intervene in situations where the interests of the major powers are not at stake, and that the UN is ineffectual without the direct involvement of US military force.

Despite the political difficulties that have surrounded UN involvement in Rwanda (1994), Kosovo (1999) and Darfur (2004), there is increasing willingness to intervene against despotic governments and to support humanitarian intervention to protect civilians. These are examples of human rights wars. What has changed historically to make human rights a prominent feature of global attempts to regulate violence? First, the globalization of communications has created opportunities for criticism of government actions, and these channels cannot be easily regulated or scrutinized by governments. The development of photography has facilitated the rapid communication of war crimes and military violence. Media coverage of the Vietnam War (1965–73) was an important turning point in the creation of global audiences of war. Newspapers such as *Al-Jazeera* and countless websites provided an alternative view of the war in Iraq (Mann 2003). Secondly, technological changes in warfare have made civilians increasingly the target of military conflict. The bombing of civilians in the Basque town of Guernica in 1937 during the Spanish Civil War (1936–9) has become the symbol of such atrocities. The carnage of the Second World War and the genocide of Jews, gypsies, homosexuals, the disabled,

Armenians, the mentally ill and so forth were important and direct causes of twentieth-century human rights legislation.

The recognition of civilian individuals rather than states as victims of war has been a major achievement of human rights legislation. The human rights movement has flourished alongside the erosion of the strong Westphalian doctrine of state sovereignty and the ascendancy of the status of the individual as a victim of war between and within states. In the aftermath of the First World War, the Allies upheld the traditional legal view that only states were the legitimate subjects of international law. The recognition of the responsibility of states towards individuals has underpinned the development of theories of reparations, that is of making good again (*Wiedergutmachung*). For example, rape has been historically regarded as simply an inevitable outcome of war, but in 2001 the International Tribunal for the former Yugoslavia found three Serbian soldiers guilty of rape as a crime against humanity. The development of the sovereignty of individuals and the responsibility of states to maintain the well-being of citizens are two important consequences of globalization. While states may regard rape in war as a crime, the sexual abuse of women and children is a common pattern of new wars.

BIO-ECONOMIC GLOBALIZATION: INFECTIOUS DISEASE AND ORGAN TRANSPLANTS

By the 1970s it was assumed that the conquest of disease in Western societies would require the development of drugs which would delay or manage old age. As medical attention moved from acute to chronic disease, preventative medicine and health education came to focus on the containment of such conditions as diabetes, stroke and heart disease, which were seen to be related more to lifestyle rather than to genetic legacy. These assumptions were severely challenged in the 1980s by the emergence of the HIV/AIDS epidemic. First reported in 1979, HIV spread rapidly among the gay and homosexual communities of North America, Europe and Australia, and subsequently to heterosexual couples and finally to drug users who shared needles. AIDS is simply one illustration of a dramatic globalization of disease.

In previous centuries, while plagues were spread by migration and trade, diseases remained relatively specific to geographical niches. With the growth of world tourism and trade, the global risk of infectious disease has increased rapidly. Global influenza epidemics now spread almost instantaneously. There is widespread anxiety about the development of a variety of new conditions that are difficult to diagnose and classify, complex in their functions and diffusion, and resistant to conventional medical treatment. The list of such conditions now includes the eruption of newly discovered diseases such as hantavirus, the migration of diseases to new areas such as cholera in Latin America, diseases produced by new technologies such as toxic shock syndrome and Legionnaires' Disease, and diseases which spring from animals to humans such as AIDS and SARS. A unique form of pneumonia, SARS is caused by an influenza-type virus rather than by bacteria. The World Health Organization warned that SARS was moving across the world with 'the speed of a jet'. By early April 2003, there were 1,268 reported cases and 61 deaths in Hong Kong. Having

infected hundreds of people in China and Vietnam, the condition spread quickly to North America and Europe, being carried mainly by airline passengers who were leaving Asia through Singapore and Hong Kong. SARS had subsequently a major impact on the Asian economy and led many air carriers in the region to cancel flights as governments warned their citizens not to travel. Similarly in 1999 the West Nile Virus (WNV) arrived in New York and the epidemic has been growing as it moves westwards. In 2002 the virus caused 4,156 confirmed cases and 284 deaths. WNV was expected to arrive in the densely populated areas of California by 2004, where mosquitoes are efficient carriers of the infection.

These problems have, along with Ebola, Marburg virus, Lassa fever and swine flu, generated a concern for 'the coming plagues' as a consequence of globalization (Garrett 1995). Many health agencies are now predicting that a world pandemic is highly likely in the event of avian flu spreading through the human population within the next decade and that such an outbreak would kill between 180 and 300 million people (Garrett 2005). If the current H5N1 strain produced a pandemic on the same scale as the 1918–19 flu outbreak, then approximately 1.7 million Americans would die. There is currently little protection against such an outbreak because there is a scarcity of relevant vaccines and the world's public health systems have long been in decline, because neoliberal policies have encouraged private insurance not public health. The economic consequences and social disruption of a pandemic would be enormous. The crisis resulting from a pandemic would require determined and concerted international cooperation, but there is currently little preparedness for the social damage that a pandemic would create. The globalization of infectious disease is produced by the rapid transportation of people and goods, population density and growing dependency on intensive livestock production. In particular, the transportation of animals and animal products has created ideal conditions for the transport of fungi, viruses and bacteria that produce devastating sickness in both human and animal populations (Karesh and Cook 2005).

Modern society is often described as a medical-industrial complex, because the human body can be broken down into parts that in turn become commodities. These social changes have their specific roots in commercial applications of genetic research, but perhaps a more dramatic and emotive illustration of commercialization is the emergence of a world market in kidney sales from poor to rich countries. It is controversial because its supporters argue that this organ exchange provides immediate cash benefits for the donating family while the sick receive a much-needed organ donation. Critics argue that organs typically come from women of poor countries who donate them to men in rich societies. These women do not receive adequate medical support after donation and often suffer from infections and exhaustion, and the costs of drugs to suppress the immune system of recipients represents a significant economic burden on recipient families.

Organ donation is in formal terms voluntary, but in practice kidney donation is a response to extreme poverty. Impoverished Indian women are often forced to sell a kidney in order to feed their families, but donations have not always been voluntary or with the patient's consent. Before the end of South African Apartheid, human tissues and organs were harvested from people in intensive care units without the knowledge of their families (Scheper-Hughes 2001a, 2001b). These practices raise difficult moral and political questions, because families that donate organs clearly

need the injection of cash that such transactions produce, but the exchanges are typically unequal. In the future, kidney disease will more likely be treated by therapeutic cloning.

Global markets in organ sales have been of considerable interest to medical sociologists, who have generally been critical of organ sales and bio-harvesting from remote and isolated aboriginal communities. Given the social inequalities of the existing market, therapeutic cloning would be a significant medical advance and would probably undermine the market for human organs. In practice, such medical improvements in the developed world are unlikely to help Third World societies until more fundamental social and economic improvements take place.

The tensions between legal and economic regulation are particularly problematic in the control of the sale of human organs. The potential marketization of women's bodies by reproductive technologies was early recognized by feminist critics, but Internet marketing of surrogate mothers and human gametes is now a global phenomenon. The technology of the Internet promotes the globalization of the marketing of the human body, which in turn was made possible through new reproductive technologies, and presents the greatest challenge to any attempt to regulate the market through the imposition of ethical standards upheld by law. The global market in human organs is consequently anomic. The next regulatory problem arising in this medical field is the corporate control of genetic codes via patenting. Such patents are important because they make the global sale of genetic information commercially viable, but they also insure that the economic inequalities between the developed and developing world will continue. The contemporary global conflict over stem-cell research illustrates the tensions between national politics, international regulation and corporate profits. In Singapore and South Korea, where legal regulation of stem cell and related biological research fields is permissive, there is an increasing possibility that such global scientific competition will result in the cloning of human beings. This outcome will have major moral and legal implications for what is human and posthuman (Fukuyama 2002).

BIO-ECONOMIC GLOBALIZATION: THE GERONTOLOGICAL REVOLUTION

The demographic revolution and the greying of populations are worldwide phenomena with significant consequences for the global economy, welfare policies and defence strategies. Ageing populations mean that there are significant shortages in the labour markets of the advanced societies and the future implications of ageing for investment, pensions and economic stagnation are significant. The advanced economies (in North America, Europe and Japan) have become dependent on labour migration, both legal and illegal, to solve labour shortages. However, advances in biological sciences have created an expectation that ageing could be delayed, if not cancelled. In conventional gerontology, the question about 'living forever' might in practical terms mean living a full life and achieving the average expectation of life. More recently, there has been considerable speculation as to whether medical science through stem-cell research could in fact reverse the ageing process. Between the 1960s and 1980s the view put forward by biologists was that normal cells had a

'replicative senescence', that is, normal tissues can only divide a finite number of times before entering a stage of advanced quiescence. Cells were observed *in vitro* in a process of natural senescence, and eventually experiments *in vivo* produced a distinction between normal and pathological cells in terms of division. It is paradoxical that pathological cells appear to have no necessary limitation on replication, and 'immortalization' is the distinctive feature of a pathological cell line. By extrapolation, biologists concluded that finite cell division meant that the ageing of the whole organism was inevitable. These research findings confirmed the traditional view that human life had an intrinsic and predetermined limit. Only pathology described how certain cells might out survive the otherwise inescapable senescence of normal cellular life.

This biological framework of ageing was eventually disrupted by the discovery that some isolated human embryonic cells were capable of continuous division in culture, and showed no evidence of a replicative crisis. Certain non-pathological cells (or stem cells) were capable of indefinite division, and hence were 'immortalized'. The cultivation of these cells as an experimental form of life has challenged existing assumptions about the boundaries between the normal and the pathological, and between life and death. Stem cell research begins to define the arena within the body that has reserves of renewable tissue, indicating that the limits of biological growth are not fixed or inflexible. The body has a surplus of stem cells capable of survival beyond the death of the organism. With these developments in biogerontology, the capacity of regenerative medicine to expand the 'natural' limits of life becomes a plausible aspect of contemporary medicine. This new interpretation of replication locates ageing as a shifting threshold between surplus and waste, between obsolescence and renewal, between death and immortality.

The ageing of the developed world is regarded by economists in the World Bank as a threat to economic growth, and hence there is much political excitement about the possibilities of stem cell research as an aspect of regenerative medicine. Companies operating in tourist destinations as far afield as the Caribbean and Malaysia are already offering regenerative medicine as components of holiday packages, designed to alleviate the negative consequences of degenerative diseases such as multiple sclerosis or diabetes. Holiday packages will come to include cosmetic and regenerative medicine as part of the world cruise for the super-rich. The idea of gerontological tourism offering rest, cure and rejuvenation might also become an addendum to sexual tourism in the world of advanced bio-capitalism. One sign of the times was an academic event hosted by Cambridge University Life Extension and Rejuvenation Society in October 2004 in which Dr Aubrey de Grey announced that human beings could 'live forever', by which he meant that within 25 years medical science will possess the capacity to repair all known effects of ageing. The average age at death of people born thereafter would exceed 5,000 years.

Such expectations are literally fantastic. In fact the scientific expectations of significant breakthroughs in the treatment of disease and significant profits by the large pharmaceutical companies after the decoding of the human genome in 2001 were not fulfilled. The pharmaceutical industry has been reluctant to invest in new products that were designed for conditions that affect small numbers of people. The fears associated with 'personalized medicine' have begun to disappear, because it is obvious that there are generic processes from which the genomics companies can

profit. However, genetics-based medicine is poised to find better diagnostic tests for disease and generic solutions to such conditions as diabetes, Alzheimer's disease, heart problems and breast cancer. These advances will undoubtedly enhance life expectancy and have significant implications for the shape of the world's population in terms of age structure.

The human consequences of these medical innovations and demographic changes will be rapid and radical, but little systematic thought has been given to the social and political consequences of enhanced longevity. Although it is mere speculation, it is reasonable to assume that the social outcomes of new patterns of ageing will be revolutionary. Increasing world inequality between the rejuvenated North and the naturally ageing South would further inflame frustration and resentment of deprived social groups against wealthy aged populations. Under existing economic conditions, labour markets will be unable to cope with the increasing number of ageing survivors and there would be major crises in housing markets. The food supply would be unable to keep up with population expansion and there will be increasing dependence on the use of genetically modified food. As environmental pollution increases, nation-states will have to struggle to maintain natural resources to support their own populations. In addition to national conflicts, there would be intergenerational conflicts over resources, in which existing economic crises around pensions and housing would increase. The elderly would be increasingly uninsurable and would need to work indefinitely to support themselves and their families in old age. If we assume that while the application of genetic sciences could reduce mortality, it would, at least in the short term, increase morbidity as chronic illness and geriatric diseases increased. The burden of dependency would have negative consequences for healthcare systems and economic growth. Since longevity does not in itself result in greater happiness, elderly populations would be drug-dependent. The main driving force in the economy would be from genetic research and its applications in pharmaceutical industries. A pharmaceutical product can be profitable when applied to a chronic condition which it does not cure, but where it helps the patient to survive in a relationship of complete dependency on the new drug. The aim of commercial pharmacy is not to cure but to assist survival. There are strong economic reasons to promote longevity, provided it is longevity with increasing morbidity.

The prospect of indefinite life would thus raise an acute Malthusian crisis. These changes imply an important change from early to late modernity. In the early stages of capitalism, the role of medical science was to improve healthcare to make the working class healthy in order to have an efficient labour force. Late capitalism does not need a large permanent labour force at full employment and working full time, because technology has made labour more efficient. In the new biotechnological environment, disease is no longer a negative force in the economy but on the contrary an aspect of the factors of production. The economy will increasingly depend on the demand for products to manage the chronic illness of a population condemned to 'live forever'. These assumptions about the dominance of chronic illness will be periodically undermined by pandemics that will assist in reducing the world's population pressures. The relationship between pandemics and ageing will produce uncertain and contradictory outcomes for the global economy.

This Malthusian crisis has been brilliantly conceptualized by Nicholas Georgescu-Roegen (1971) in his *The Entropy Law and the Economic Process* where he argues

that the exhaustion of the earth's resources cannot be finally overcome by economic growth, because waste is an unavoidable law of nature, and the economic process merely accelerates the realization of this law. The economic process involves an inevitable degradation and depletion of resources. Economics has strangely neglected the problem of the scarcity of natural resources and Georgescu-Roegen attempted to develop an economic science based on assumptions about biological evolution and the economic process of production and consumption as parallel and interconnected processes. Following the biologist Alfred Lotka, he noted that humans like other animals use their organs to enhance their enjoyment of life. All animals are born with these organs that Lotka called endosomatic instruments, but man is the only creature who uses organs that are not part of his biological constitution. These are his exosomatic instruments. In economic terms, these exosomatic instruments are in fact simply capital equipment. This bio-economic understanding of human beings can be combined with the philosophical anthropology of Arnold Gehlen to argue that human beings are poorly equipped with endosomatic instruments and hence have to build institutions to protect themselves and to create their social world. These exosomatic instruments are inherently precarious and the somatic condition of human beings is vulnerable. Therefore the logical situation of human beings has one major problem, namely scarcity and the resulting social conflict over resources (Turner and Rojek 2001).

Georgescu-Roegen conceptualizes this state of scarcity in terms of the production of waste in the economic process, namely the inevitable depletion of resources and the pollution of our environment that are associated with population growth. In his entropic economics, Georgescu-Roegen rejected Karl Marx's solution that the socialization of the means of production by communism could overcome the social class conflict resulting from the struggle over scarce resources, and rejected the idea that endless technological innovation can enhance our resources and reduce the expenditure of labour in relation to leisure time and consumption. In this economic interpretation of waste, the depletion of resources is inevitable, and hence social conflict results from scarcity and overpopulation. Environmental depletion has been rapid and profound. For example, in a period of some 40 years, the forest cover in Thailand was reduced from 65 per cent to 15 per cent resulting in catastrophic flooding, pollution of rivers and decline in soil quality. The proposal that we could 'live forever', even if that means an extra decade for rich nations, can only increase the rate at which resources are depleted and pollution increased, and it would inevitably lead to war.

CONCLUSION: WASTE, WAR AND WARRIORS

We can now summarize this neo-Malthusian argument by claiming that globalization by creating global interconnectedness increases the social impact of technological change. Global social change will be increasingly negative, because it will magnify entropic waste. Globalization of tourism, transport and exchange, together with increasing population density, produces environmental degradation and major epidemics that cannot be easily contained. If a pandemic is inevitable under these circumstances, there will be major economic crises and the resulting social

disruption will further enhance the possibilities of civil and military conflict. The surplus of unemployed and underemployed young men is a major causal factor in the global development of new wars, that will bring with them social breakdown. Sociologists have begun to talk about the destructive social role of the 'overpopulation warrior' (Diessenbacher 1998) whose presence has caused havoc in the Congo, Myanmar, southern Thailand, Cambodia and central Asia. Social disruption in the advanced societies is also closely related to the absence of adequate social roles for young men in economies that are outsourced and that no longer need permanent, mass labour supplies. Casualization of employment, underemployment and low status leads to lack of respect, low social capital and marginalization from citizenship institutions. The London bombers were not impoverished, but their social alienation was expressed through their sense of not being UK subjects, and hence the Islamic *ummah* offered some subjective notion of inclusion and purpose.

Environmental crisis will become an increasingly significant determinant of the economic viability of nation-states, but it is doubtful whether there are such things as 'natural disasters'. The loss of life in the tsunami of 2004 was in part a consequence of the absence of an early warning system. The flooding of New Orleans in 2005 was a consequence of failed levees and inadequate planning, but it was also a consequence of global warming, and the resulting social chaos and collapse of civil order can be taken as one illustration of the social and political disruptions that will flow from further globalization. Deforestation is clearly a man-made disaster that will cause sea levels to rise as the ice cap defrosts and sea temperatures rise. It is likely, not only that many small Pacific islands will simply disappear, but low-lying areas of Europe, such as the English fens, will be under water. These environmental changes will increase scarcity and make conflict over water and land more prominent in civil conflict. War and waste will be significant driving features of the futures of globalization.

In the past, catastrophes such as the Black Death, the fire of London and the Lisbon earthquake produced major religious responses. The very concept of theodicy in the philosophy of Leibniz attempted to explain such disasters, and was adopted by Max Weber in his analysis of religious ideologies. For example, the Asian tsunami resulted in a powerful religious response in fundamentalist circles in Indonesia, where video-tapes were soon available showing that the natural disaster was a punishment from God for human wickedness in Thailand and elsewhere. Human failure to adhere strictly to religious law (the Shari'a) was allegedly the real cause of natural catastrophe. In medieval Europe, the Black Death of the fourteenth century, which was carried by the intercontinental trade and warfare, resulted in the deaths of one-third of the population of Europe. The plague broke out in 1346 among the armies of a Mongol prince who was attacking the Crimea. Despite the limitations of trade routes, even Greenland suffered from the epidemic. The plague which became a persistent but unpredictable feature of society, brought about a new consciousness, namely a macabre sense of the persistent presence of a malevolent force bringing certain destruction, evil and death. The baroque imagination of the post-Reformation Europe sought to explain the catastrophe of Lutheranism. Comparisons with modern epidemics and their complex social and political history such as malaria are useful from an historical and sociological perspective. The impact of tuberculosis would be another example of a disease that created a new

consciousness and culture, especially among the intellectual elite (Dormandy 1999). The globalization of disease was powerfully illustrated in 1918 with the outbreak of the Spanish influenza pandemic that killed 25 million people in 6 months – about three times the number of deaths of World War I. In 1920, an epidemic of encephalitis lethargica and a further outbreak of influenza followed. These epidemics were rapid and global, because the war had brought about worldwide movements of peoples. As a consequence influenza spread rapidly among US troops based in Europe, and between September and October 1918, 20 per cent of the US army was sick and 24,000 soldiers died of flu compared to 34,000 killed in battle. In the 1920s in response to the mass slaughter of young men in the trenches, there was a significant expansion of interest in spiritualism Europe in which figures like Conan Doyle became involved in the collective attempt to reach the departed. Wartime horror produced a widespread need to communicate with the dead, especially where grieving relatives could not locate the bodies of their loved ones.

From these examples, it is evident that there is no clear point that divides global history between simple and reflexive modernity, between stable and mobile societies. Popular images or metaphors of AIDS are not far removed from the baroque vision of death (Ziegler 1998). While there are forms of social continuity, globalization will make catastrophes more general, immediate and profound. The pandemics of the future, precisely because the world is interconnected but lacks any effective global governance, will destroy millions of people, undermine whole societies and threaten civilized life. If the current weakness of the United Nations continues, it is unlikely that any single society, such as the United States, could act as a global policeman, or a global doctor. In the context of uncontrollable pandemics and famines, we can expect millenarian movements and fantastic cults to flourish once more in the era of global catastrophes. In this respect, the future of globalization will resemble not the *Brave New World* of Huxley or that of Orwell in *Nineteen Eighty-Four*, but the world of Mad Max with roaming armies of displaced men in search of gasoline, armaments and drugs, where small fortified hamlets of the rich and powerful would seek shelter, and secure religious meaning from their cults and prophets. This future world is already coming into existence in Kinshasa, Kabul, Baghdad, Manipur, the Fergana valley and Grozny, where the ingredients of the global neo-Malthusian crisis – infectious disease, alienated men, drug dependency, civil disruption, environmental decay, control by war lords and new wars, and religio-political extremism – are being unwittingly assembled. These new-war conflict zones are the outposts of empire that will drain away the resources of democratic societies and erode the civil liberties of their citizens. It is unlikely that the juridical revolution can contain such pressing global anarchy.

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